

Name: _____

Class: _____

Date: _____

Unit 4 Review Packet

Computer Systems & Networks



How to use this packet

Match the vocabulary, then work the mixed review questions. These are the exact words and question types you'll see on the unit test and the AP exam.

Vocab Match 1

_____ 1. Internet

_____ 2. Protocol

_____ 3. Bandwidth

_____ 4. Router

_____ 5. World Wide Web

_____ 6. Fault tolerance

_____ 7. Redundancy

_____ 8. Single point of failure

_____ 9. Path

_____ 10. Reliability

A. a network of interconnected networks that use standardized, open (nonproprietary) protocols

B. how consistently a system works correctly

C. a system's ability to keep working when some parts fail

D. one part whose failure disconnects or stops the system

E. a device that forwards packets toward their destination

F. extra components or paths used as backups

G. a sequence of directly connected devices from sender to receiver

H. an agreed-upon set of rules that specify the behavior of a system

I. a system of linked pages, programs, and files that uses the Internet

J. the maximum amount of data sent in a fixed amount of time, usually in bits per second

Vocab Match 2

_____ 1. Sequential computing

_____ 2. Parallel computing

_____ 3. Distributed computing

_____ 4. Speedup

A. sequential time divided by parallel time

B. some steps run at the same time on multiple processors

C. a program runs across multiple devices (computers)

D. steps are run one at a time, in order

Unit 4 Review Packet

Mixed Review

Mixed Review

1. Three independent tasks take 10 s, 15 s, and 20 s. On THREE processors, how long does the whole job take?

- A. 10 s
- B. 20 s
- C. 15 s
- D. 45 s

2. A program takes 100 s on one processor. 20 s must run sequentially; the other 80 s can be split evenly among processors. How long on 4 processors?

- A. 25 s
- B. 80 s
- C. 20 s
- D. 40 s

3. Which TWO statements are true? (*Select two.*)

- A. Redundancy always lowers costs.
- B. Fault-tolerant systems can keep working when some components fail.
- C. Adding paths between devices can increase fault tolerance.
- D. Fault-tolerant networks never lose packets.

4. What is redundancy in a network?

- A. Sending the same message to everyone
- B. Compressing data
- C. Removing unnecessary devices
- D. Including extra components or paths that can be used if others fail

5. In the network shown on page 1, which connection, if added, would ensure that no single device failure could disconnect A from F?

- A. A-D
- B. B-E
- C. B-C
- D. C-D

6. Which TWO statements about packets are true? (*Select two.*)

- A. All packets from one message travel the same path.
- B. Packets can arrive out of order.
- C. Packets contain metadata used for routing.
- D. Packets are only used to send email.

7. Why is the Internet considered fault-tolerant?

- A. Its devices never fail
- B. It uses only wireless connections
- C. Every packet takes the same route
- D. Data can be rerouted along other paths when a device or connection fails

8. What is a protocol?

- A. An agreed-upon set of rules that specify the behavior of a system
- B. A type of computer hardware
- C. A program that blocks viruses
- D. The fastest path through a network

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9. Which describes **parallel** computing?

- A. Operations run one at a time in order
- B. A program runs only on a single core
- C. Data is compressed before it's sent
- D. A program is broken into smaller operations, some of which run at the same time

10. Four independent processes take 60, 30, 50, and 40 seconds. What is the minimum time on TWO processors?

- A. 60 s
- B. 90 s
- C. 100 s
- D. 180 s

11. Why has the Internet's use of open protocols been important to its growth?

- A. Any device or company can follow them to connect and communicate
- B. Open protocols encrypt all data automatically
- C. They make the Internet owned by one company
- D. They prevent packets from being lost

12. Which best describes the relationship between the Internet and the World Wide Web?

- A. They are the same thing
- B. The Internet is part of the Web
- C. The Web is a system of linked pages and files that uses the Internet
- D. The Web was created before the Internet

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Challenge Questions

Challenge 4.1

Your home Internet has a bandwidth of 100 megabits per second. Ideally, how long does it take to download a 500 megabyte file? Show your work and explain why bits vs. bytes matters here.

Challenge 4.2

A school network has only ONE connection to the Internet. Explain why this is not fault-tolerant, propose an improvement, and describe one trade-off of your improvement.

Challenge 4.3

Using question 4: what is the shortest possible time with unlimited processors, and the maximum speedup? Explain why adding more processors eventually stops helping.
